

Procedure: Attaching a Role to an EC2 Instance

The screenshot displays the AWS IAM Dashboard. On the left is a navigation sidebar with the following sections:

- Identity and Access Management (IAM)**
 - Search IAM
 - Dashboard
 - Access Management
 - User groups
 - Users
 - Roles**
 - Policies
 - Identity providers
 - Account settings
 - Root access management
 - Temporary delegation requests
 - Access reports
 - Access Analyzer
 - Resource analysis

The main content area is titled **IAM Dashboard** and includes a blue banner at the top for **New! IAM Outbound Identity Federation**. Below the banner, the **Security recommendations** section shows two items:

- Root user has MFA**: Having multi-factor authentication (MFA) for the root user improves security for this account.
- Root user has no active access keys**: Using access keys attached to an IAM user instead of the root user improves security.

The **IAM resources** section provides a table of resources in the AWS Account:

IAM resources				
Resources in this AWS Account				
User groups	Users	Roles	Policies	Identity providers

On the right side of the dashboard, the **AWS Account** section displays:

- Account ID**: 848186563482
- Account Alias**: [Create](#)
- Sign-in URL for IAM users in this account**: <https://848186563482.signin.aws.amazon.com/console>

At the bottom right, the **Quick Links** section includes a link for [My security credentials](#).

Create the Role: Go to IAM > Roles > Create role

IAM > Roles

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Users groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource access

Roles (96) Info

Delete

Create role

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

< 1 2 3 4 5 >

☐	Role name	Trusted entities	Last activity
☐	23-03-26	AWS Service: ec2	-
☐	26-03-26	AWS Service: ec2	11 days ago
☐	AmazonSageMaker-ExecutionRole-20230505T100707	AWS Service: sagemaker	997 days ago
☐	aws-elasticbeanstalk-service-role	AWS Service: elasticbeanstalk	-
☐	awsrek-role-kwvnue3a	AWS Service: lambda	-
☐	AWSServiceRoleForAmazonElasticFileSystem	AWS Service: elasticfilesystem (Servi	106 days ago
☐	AWSServiceRoleForApplicationAutoScaling_DynamoDBTable	AWS Service: dynamodb.application	706 days ago
☐	AWSServiceRoleForBackup	AWS Service: backup (Service-Linked	21 hours ago

Select Trusted Entity: Choose AWS Service.

☰ IAM > Roles > Create role

ⓘ

Step 1

☒ Select trusted entity

Step 2

☐ Add permissions

Step 3

☐ Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").

[IAM](#) > [Roles](#) > Create role

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ EC2

Allows EC2 instances to call AWS services on your behalf.

☐ EC2 Role for AWS Systems Manager

Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.

☐ EC2 Spot Fleet Role

Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.

☐ EC2 - Spot Fleet Auto Scaling

Allows Auto Scaling to access and update EC2 spot fleets on your behalf.

☐ EC2 - Spot Fleet Tagging

Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.

☐ EC2 - Spot Instances

Allows EC2 Spot Instances to launch and manage spot instances on your behalf.

Add Permissions: Search for and check AmazonS3FullAccess.

☰

[IAM](#) > [Roles](#) > Create role

ⓘ

Step 1
● Select trusted entity

Step 2
● **Add permissions**

Step 3
○ Name, review, and create

Add permissions [Info](#)

Permissions policies (1/1169) [Info](#)

ⓘ

Choose one or more policies to attach to your new role.

Q S3

×

Filter by Type

All types ▼

26 matches

< 1 2 > ⓘ

☐	Policy name ↗	Type	Description
☐	☐  AmazonDMSRedshiftS3Role	AWS managed	Provides access to manage S3 settings ...
☐	☐  AmazonS3ExpressFullAccess	AWS managed	Provides full access to all S3 directory ...
☐	☐  AmazonS3ExpressReadOnlyAccess	AWS managed	Provides read only access to S3Express...
☒	☒  AmazonS3FullAccess	AWS managed	Provides full access to all buckets via t...
☐	☐  AmazonS3ObjectLambdaExecutionRolePolicy	AWS managed	Provides AWS Lambda functions permi...
☐	☐  AmazonS3OutpostsFullAccess	AWS managed	Provides full access to Amazon S3 on ...

Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.

The screenshot shows the AWS IAM console interface for creating a new role. The breadcrumb navigation at the top indicates the path: IAM > Roles > Create role. On the left, a vertical sidebar shows the three steps of the process: Step 1: Select trusted entity, Step 2: Add permissions, and Step 3: Name, review, and create. Step 3 is currently selected and highlighted with a blue circle. The main content area is titled 'Name, review, and create'. It contains two sections: 'Role details' and 'Step 1: Select trusted entities'. The 'Role details' section has a 'Role name' field with the text 'CC-Role-Demo' and a 'Description' field with the text 'Allows EC2 instances to call AWS services on your behalf.' Below these fields are instructions on character limits and allowed characters. The 'Step 1: Select trusted entities' section has an 'Edit' button and a 'Trust policy' section containing a JSON snippet for a trust policy.

IAM > Roles > Create role

Step 1
● Select trusted entity

Step 2
● Add permissions

Step 3
● **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

CC-Role-Demo

Maximum 64 characters. Use alphanumeric and '+, @, _' characters.

Description
Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+@-/\[\]#\$%^&*~'

Step 1: Select trusted entities [Edit](#)

Trust policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [
```

Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.

☰ IAM > Roles > Create role ⓘ

13
14
15
16

Step 2: Add permissions Edit

Permissions policy summary

Policy name ↗	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional ⓘ

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Previous Create role

Role has been created



Role CC-Role-Demo created.

View role

DeleteCreate role

Roles (97) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

X3 matches<1>

☐	Role name	Trusted entities	Last activity
☐	CC-Role-Demo	AWS Service: ec2	-
☐	dynamodbfull--access	AWS Service: lambda	17 days ago
☐	s3--full--access	AWS Service: lambda	 706 days ago

Roles Anywhere [Info](#)

Manage

Authenticate your non AWS workloads and securely provide access to AWS services.



Access AWS from your non AWS workloads

Create your own AWS workload using the same...



X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to...



Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

Create the ec2 instance and and connect to it and check for aws ec2 describe-instances

Attach to Instance:

Go to the EC2 Dashboard. Select your running instance.

Click Actions> Security> Modify IAM role.

EC2 > Instances

Instances (1/1) Info

Last updated 6 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☒	Name	Instance ID	Instance state
☒	Role-demo	i-0b711793773d1361b	Running

Change security groups
Modify IAM role
Get Windows password

Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot

i-0b711793773d1361b (Role-demo)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-0b711793773d1361b	Public IPv4 address 18.216.91.107 open address	Private IPv4 addresses 172.31.30.145
IPv6 address	Instance state	Public DNS

Select the role you just created and click Update IAM role

EC2 > Instances > i-0b711793773d1361b > Modify IAM role

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID

 i-0b711793773d1361b (Role-demo)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

CC-Role-Demo ▼

 [Create new IAM role](#)

[Cancel](#) [Update IAM role](#)

Verify via CLI:

SSH/Connect into your EC2 instance. Type `aws s3 ls`.

The Result: The command works immediately.

We can here role has only s3 full access only here-- `aws ec2 describe-instances` – unauthorized

You did not have to provide an Access Key or Secret Key. AWS automatically provided

Temporary Security Credentials to the instance

```
[ubuntu@ip-172-31-40-202:~$ aws s3 ls
[ubuntu@ip-172-31-40-202:~$ aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:sts::904233099191:assumed-role/EC2S3Role/i-0d294b9e154a4cf15 is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
ubuntu@ip-172-31-40-202:~$ █
```

Scenario: New Developer Access

You hired a new developer who needs access to EC2 and S3.

Question: How will you provide access?

Ans: Create an IAM user and attach policies granting EC2 and S3 access.

- Scenario: Temporary Access Needed

A user needs access only for 2 days. How to do this?

Ans: Use an IAM role with temporary credentials via STS for 2-day access.

- Scenario: Access Key Leak

A developer accidentally exposed access keys on GitHub. How to resolve this?

Ans: Immediately deactivate/delete keys, rotate credentials, and review logs.

- Scenario: CLI Access Required

User needs programmatic access. How to do this?

Ans: Create an IAM user and generate access key and secret key for CLI use

- Scenario: 50 Employees Same Permissions

50 employees need same S3 access. How to do this?

Ans: Create an IAM group, assign S3 policy, and add all 50 users.

- Scenario: Change Permissions for Multiple Users

You need to revoke access from 20 users. How to do this?

Ans: Remove or update permissions at group level or detach policies from users

- Scenario: Different Departments

HR, Dev, Admin need different access. How to do this?

Ans: Create separate IAM groups (HR, Dev, Admin) and assign specific policies.

Scenario: EC2 Access to S3

An EC2 instance needs access to S3 without credentials. How to do this?

Ans: Attach an IAM role with S3 permissions to the EC2 instance.

- Scenario: Cross-Account Access

Account A wants to access resources in Account B. How to do this?

Ans: Create an IAM role in Account B with trust policy for Account A.

- Scenario: Lambda Access to DynamoDB

Lambda needs to read/write DynamoDB. How to give this?

Ans: Assign an IAM role to Lambda with DynamoDB read/write permissions.

- Scenario: Temporary Elevated Access

Admin access needed temporarily. How to do this?

Ans: Use an IAM role with admin access and allow temporary role assumption.

- [Scenario: Restrict S3 Access to Specific Bucket](#)

User should only access one bucket. How to do this?

Ans: Attach a policy restricting access to only the specific S3 bucket ARN.

- [Scenario: Deny Delete Actions](#)

Users should not delete resources. Why this happens?

Ans: Because an explicit Deny in IAM policy overrides all Allow permissions.

- [Scenario: Time-Based Access](#)

User should access only during office hours. How to do this?

Ans: Use IAM policy with time-based condition (aws:CurrentTime).

- [Scenario: IP-Based Access](#)

Allow access only from office network. How to do?

Ans: Use IAM policy with IP address condition restriction.

- Scenario: Secure Best Practice

How do you manage permissions efficiently?

Ans: Follow least privilege principle and use IAM roles and groups.

- Scenario: Root User Usage

When should root user be used?

- Scenario: Multiple Services Access

User needs EC2 + S3 + DynamoDB. How to do this?

Ans: Use root only for critical account setup and billing tasks.

- Scenario: Audit Permissions

How to track user activity?

Ans: Enable CloudTrail to log and monitor all activities.

- Scenario: Over-Permission Issue

User has too much permission. How to solve this?

Ans: Reduce permissions using least privilege and IAM Access Analyzer.

